

3(a)	sum of CW moments = sum of ACW moments	M1
	about the same point for (an object in rotational) equilibrium	A1
3(b)	moment = $0.3(0) \times 0.29 \cos 40^\circ$ or $0.3(0) \times 0.222$	C1
	= 0.067 N m	A1
3(c)(i)	$k = F/x$ or $k = \text{gradient}$	C1
	e.g. $k = 21/10 \times 10^{-3}$ $k = 2100 \text{ N m}^{-1}$	A1
3(c)(ii)	$V_{\text{sphere}} = \frac{4}{3} \times \pi \times (0.0480)^3$	C1
	$F = \rho g V$ (upthrust =) $1000 \times 9.81 \times \left(\frac{4}{3} \times \pi \times (0.048)^3\right) \times 0.26(0) = 1.18 \text{ (N)}$	A1
3(c)(iii)	1.18×0.29 or 0.30×0.29 or $F \times 0.017$	C1
	$(1.18 \times 0.29) = (0.30 \times 0.29) + (F \times 0.017)$ $F = 15 \text{ N}$	A1

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Question	Answer	Marks
3(c)(iv)	$E_{(P)} = \frac{1}{2}kx^2$ or $E_{(P)} = \frac{1}{2}Fx$	C1
	$x = F/k = 15/2100$ or x determined from graph for $F = 15.0$ N	A1
	$E_P = \frac{1}{2} \times 2100 \times (15/2100)^2$ or $E_P = \frac{1}{2} \times 15 \times (15/2100)$	
	$E_P = 0.054$ J	
3(d)	the <u>sphere</u> has gained gravitational potential energy	B1